Supplementary Material

Norovirus detection in wastewater and its correlation with human gastroenteritis: a systematic review and meta-analysis

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Section1 Research Assessment Scale

Each included study was independently reviewed and assessed by two researchers according to the AHRQ cross-sectional study assessment scale. We modified some of the items with the following information to accommodate the included articles:

(1) define the source of information (survey, record review);

(2) list selection criteria for sampling sites or refer to previous publications;

(3) set up a control (positive control or negative control) in the method;

(4) continuous sampling or sampling time over one year and provide detailed data;

(5) describe the main processes and methods of sample processing and analysis;

(6) describe any assessments undertaken for quality assurance purpose (e.g., test/retest of primary outcome measurements, or follow MIQE guidelines for sample concentration, RNA extraction and reverse transcription);

(7) perform typing of the virus genotype and evaluate the results;

(8) describe how inhibition was assessed and/or controlled;

(9) if applicable, explain how missing data were handled in the analysis/set the LOD;

(10) evaluate the recovery rate of samples and completeness of data collection;

(11) reassess the accuracy of the results if there was an effect of the recovery rate.

Similarly, an item would be scored ‘0’ if it was answered ‘NO’ or ‘UNCLEAR’; if it was answered ‘YES’, then the item scored ‘1’. Article quality was categorised as follows: low quality = 0-4; moderate quality = 5-8; and high quality = 9-11.

Table S1 Results of study quality assessment of included 46 studies*

	Source of information	Selection criteria	Control	Sampling and detailed data	Processes and methods	Assessments undertaken	Type the virus genotype	Inhibition assessment/ controlled	Handle missing data/ set the limit of detection	Evaluate recovery rate	Reassess the result accuracy	Total score	Quality
H. Wang 2020	Yes	Yes	No	Yes	Yes	No	No	Unclear	No	No	No	5	Moderate
G. La Rosa 2008	Yes	No	No	No	Yes	No	Yes	Yes	Unclear	No	No	4	Low
S. Kazama 2016	Yes	Yes	No	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	9	High
A. H. Kamel 2010	Yes	Yes	No	Yes	Yes	No	Yes	No	No	No	No	5	Moderate
J. M. Fioretti 2018	Yes	Yes	No	Unclear	Yes	Yes	Yes	No	No	No	No	5	Moderate
A. Carducci 2006	Yes	Unclear	No	Yes	Yes	No	No	No	No	No	No	3	Low
N. Zhou 2016	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Unclear	No	7	Moderate
M. Victoria 2016	Yes	Yes	Yes	Yes	Yes	Unclear	Yes	No	No	No	No	6	Moderate
M. Victoria 2010	Yes	Unclear	Yes	Yes	Yes	Yes	Yes	No	No	Yes	No	7	Moderate
D. M. Teixeira 2016	Yes	No	Yes	Unclear	Yes	Yes	Yes	No	No	No	No	5	Moderate
D. M. Teixeira 2017	Yes	Yes	Yes	Unclear	Yes	Yes	Yes	No	No	No	No	6	Moderate
Z. Tao 2015	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	7	Moderate
E. Suffredini 2018	Yes	Unclear	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	7	Moderate
S. Skraber 2011	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	9	High
T. Prado 2011	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	7	Moderate
T. Prado 2019	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Unclear	10	High
M. Muscillo 2013	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Unclear	9	High

N. Montazeri 2015	Yes	Yes	Yes	No	Yes	Unclear	Yes	No	Yes	No	No	6	Moderate
Y. Masago 2016	Yes	Unclear	Unclear	No	Yes	Yes	Unclear	No	No	Yes	Unclear	4	Low
V. V. Mabasa 2018	Yes	Yes	No	Unclear	Yes	Yes	Yes	Yes	No	Yes	Yes	8	Moderate
G. La Rosa 2010	Yes	Yes	Yes	No	Yes	Yes	Yes	Yes	No	No	Unclear	7	Moderate
M. Kitajima 2012	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	8	Moderate
S. Kazama 2017	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	Yes	10	High
M. Hassine-Zaafrane 2014	Yes	Yes	No	No	Yes	No	Yes	No	No	No	No	4	Low
E. Haramoto 2006	Yes	Yes	No	No	Yes	No	Unclear	No	No	Yes	No	4	Low
T. H. Han 2014	Yes	Yes	No	Yes	Yes	No	Yes	No	No	No	No	5	Moderate
T. M. Fumian 2019	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	No	9	High
N. Kobayashi 2017	Yes	Yes	No	Yes	Yes	Yes	No	No	No	Yes	Yes	7	Moderate
K. Gurung 2017	Yes	Unclear	No	No	Unclear	Unclear	No	Unclear	Unclear	No	No	1	Low
C. J. A. Campos 2016	Yes	Yes	No	Yes	Yes	Yes	No	No	No	Yes	Yes	7	Moderate
M. Amarasiri 2018	Yes	Unclear	Unclear	No	Yes	Yes	Yes	Unclear	Yes	No	No	5	Moderate
U. Pérez-Sautu 2012	Yes	Yes	No	Yes	Yes	Unclear	Yes	No	Yes	No	No	6	Moderate
L. C. Sima 2011	Yes	Yes	Yes	No	Yes	No	No	Yes	Yes	No	No	6	Moderate
T. Miura, S 2015	Yes	No	Yes	Yes	Yes	No	No	Unclear	Yes	Yes	No	6	Moderate
F. G. Masclaux 2013	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes	Yes	No	8	Moderate
A. Kauppinen 2014	Yes	Yes	No	Yes	Yes	No	No	No	Unclear	No	No	4	Low
M. Hellmér 2014	Yes	Yes	No	No	Yes	No	No	Unclear	No	Yes	Yes	5	Moderate
R. C. G-R 2014	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	Yes	Yes	8	Moderate
A. K. Da Silva 2007	Yes	Yes	Yes	Yes	Yes	Unclear	No	Yes	Yes	Yes	No	8	Moderate
T. G. Aw 2010	Yes	Yes	Yes	No	Yes	No	Yes	No	No	No	No	5	Moderate
J. Lu 2021	Yes	Yes	No	Yes	Yes	Yes	Yes	No	Yes	No	No	7	Moderate

C. McCall 2020	Yes	Yes	Yes	No	Yes	Yes	No	No	Unclear	No	No	5	Moderate
X. Lin 2021	Yes	Yes	Unclear	Yes	Yes	No	Yes	No	No	No	No	5	Moderate
C. Santiso-Bellón 2020	Yes	Yes	Unclear	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	9	High
C. Ibrahim 2020	Yes	Yes	Yes	No	Yes	Yes	No	No	No	No	No	5	Moderate
J. H. Lun 2018	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	7	Moderate

*: Each included study was reviewed and assessed according to the AHRQ cross-sectional study assessment scale. We modified some of the items to accommodate the included articles. An item would be scored ‘0’ if it was answered ‘No’ or ‘Unclear’; if it was answered ‘Yes’, then the item scored ‘1’. Article quality was assessed as follows: low quality = 0~4; moderate quality = 5~8; high quality = 9~11.

Table S2 Subgroup analysis of NoV detection rate

Variable		Event	n	Detection rate(%)	95% CI	I^2(%)
Genogroup	Nov GI	1109	1748	81.90	71.26, 90.70	95.9
	NoV GII	1372	1894	88.04	77.23, 96.05	97.0
	NoV GIV	129	498	30.89	15.45, 48.73	90.8
Country Mortality Stratum	High-mortality developing	314	1018	47.08	27.85, 66.75	97.2
	Low-mortality developing	681	954	78.01	63.77, 89.68	95.5
	Developed	1615	2168	87.33	79.22, 93.87	95.4
PCR method	qRT-PCR	1852	2586	85.00	76.93, 91.74	96.8
	RT-PCR	758	1544	74.00	56.36, 88.55	97.9
Sampling duration	<1 year	481	684	81.74	67.17, 93.10	94.1
	1 year~	1581	2359	83.75	74.72, 91.26	96.2
	3 year~	548	1097	70.77	35.19, 96.21	99.2
Virus Concetration Method	Adsorption-extraction	572	786	83.79	70.11, 94.23	94.7
	PEG precipitation	788	1005	85.08	72.96, 94.35	94.8
	Ultrafiltration	166	182	91.75	85.27, 96.71	35.6
	ESMP	179	248	82.08	58.76, 97.65	93.0
	Ultracentrifugation	661	1431	72.02	52.82, 87.94	98.0
	Other method	244	488	82.80	25.16, 100.00	99.3
Toatl		2610	4140	**82.10**	74.22, 88.92	97.0

A total of 4,140 data points were obtained for the 46 articles included, of which 2,610 data points were positive for NoV detection in wastewater. According to the results of the meta-analysis, the overall detection rate was 82.10% (95% confidence interval [CI]: 74.22%-89.92%). Heterogeneity remained after the subgroup analysis ($I^2 > 50\%$) and further analysis of the source of heterogeneity is needed.

Table S3 The result of meta-regression

Covariate		Coefficient	95% CI	*P*	R^2 (%)
Univariate analysis					
Genogroup	NoV GI	-	-	-	12.00
	NoV GII	0.077	-0.196, 0.254	0.396	
	NoV GIV	-0.519	-0.858, -0.179	0.003	
Country Mortality Stratum	High-mortality developing	-	-	-	24.59
	Low-mortality developing	0.310	0.030, 0.591	0.030	
	Developed	0.420	0.160, 0.679	0.002	
PCR	qRT-PCR	-	-	-	7.96
	RT-PCR	-0.126	-0.319,0.067	0.201	
Duration	<1 year	-	-	-	0.00
	1 year~	0.0285	-0.186,0.242	0.794	
	3 year~	-0.1148	-0.456,0.227	0.510	
Virus Concetration Method	Adsorption-extraction	-	-	-	9.87
	PEG precipitation	0.02	-0.211,0.251	0.866	
	Ultrafiltration	0.087	-0.266,0.439	0.630	
	ESMP	-0.023	-0.339,0.292	0.885	
	Ultracentrifugation	-0.13	-0.368,0.108	0.285	
	Other method	-0.003	-0.414,0.409	0.990	
Multivariate analysis					52.02
Genogroup	GI	-	-	-	-
	GII	0.065	-0.065, 0.206	0.324	
	GIV	-0.552	-0.801, -0.302	<0.0001	
Country Income Level	LMIC	-	-	-	-
	UMIC	0.390	0.161, 0.618	0.0008	
	HIC	0.458	0.247,0.667	<0.0001	

* "-": The first line of covariate represent reference.

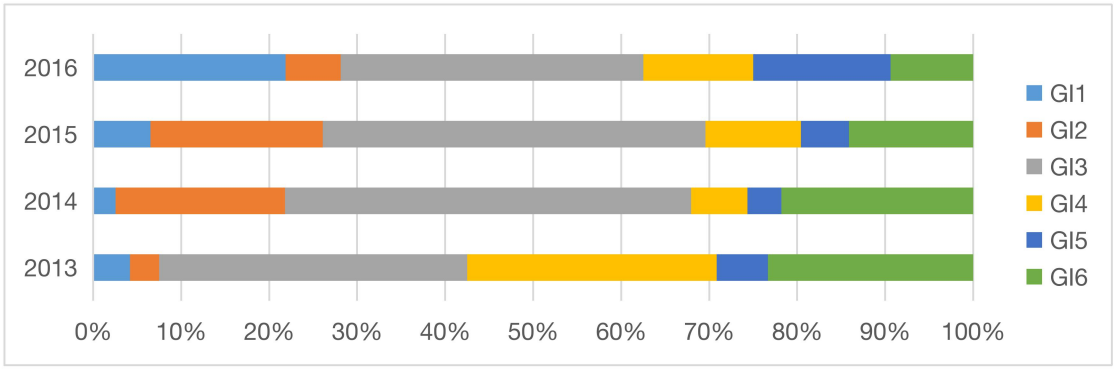
Table S4 Summary statistics for seasonal variation of NoV GI and GII concentration in wastewater (lg GC/L)

Genogroup		N	Mean	SD	95% Confidence Interval for Mean		F	P
					Lower Bound	Upper Bound		
NoV GI	Spring	155	5.56	1.89	5.26	5.85	6.459	0.000
	Summer	122	4.89	1.83	4.57	5.22		
	Autumn	124	5.00	1.82	4.68	5.33		
	Winter	175	5.69	1.91	5.41	5.98		
	Total	576	5.34	1.89	5.18	5.49		
NoV GII	Spring	162	6.15	1.42	5.93	6.37	24.868	0.000
	Summer	125	5.20	1.43	4.94	5.45		
	Autumn	133	5.08	1.56	4.81	5.35		
	Winter	191	6.21	1.52	6.00	6.43		
	Total	611	5.74	1.57	5.62	5.87		

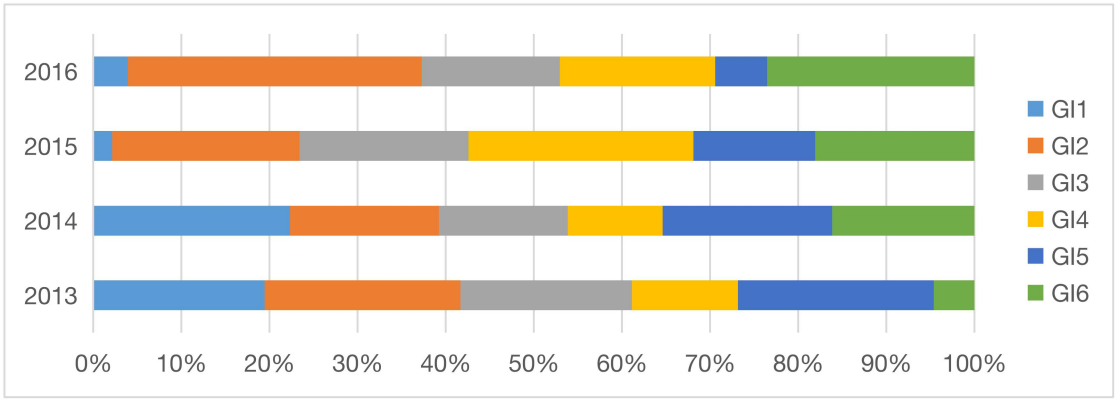
Table S5 AGE indicators (number of GI/GII AGE cases, total AGE cases and outbreaks) in local populations monthly of seven including articles

GI/GII AGE cases		Month												Total
genogroup		Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.	
GI	H. Wang 2020	4	3	1	0	1	2	0	1	1	1	2	5	21
	A. Carducci 2006	1	0	0	0	1	2	1	0	1	1	0	0	7
	M. Hellmér 2014	0	0	10	2	0	0	0	0	0	0	0	0	12
Mean		2	1	4	1	1	1	0	0	1	1	1	2	13
GII	H. Wang 2020	62	49	67	63	57	3	7	8	8	9	20	65	418
	A. Carducci 2006	2	0	0	0	1	3	2	0	0	1	0	1	10
	M. Hellmér 2014	20	83	171	42	0	0	0	0	0	0	0	0	316
Mean		28	44	79	35	19	2	3	3	3	3	7	22	248
AGE cases														
H. Wang 2020		103	99	133	111	101	25	22	23	27	21	32	101	798
S. Kazama 2016		46	23	22	0	0	0	0	0	0	0	32	42	165
S. Kazama 2017		1533	661	355	799	852	553	244	280	214	273	673	927	7364
M. Hellmér 2014		90	257	481	142	0	0	0	0	0	0	0	0	970
Mean		443	260	248	263	238	145	67	76	60	74	184	268	2324
Outbreak														
C. J. A. Campos 2016		17	12	11	6	6	5	1	1	0	0	0	13	72
J. Lu 2021		35	63	51	17	10	2	4	1	19	18	48	62	330
Mean		26	38	31	12	8	4	3	1	10	9	24	38	201

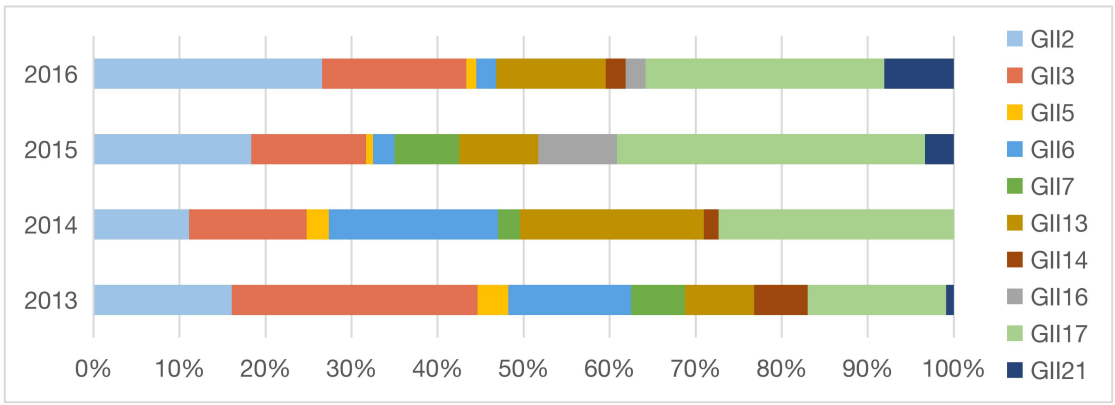
(a)

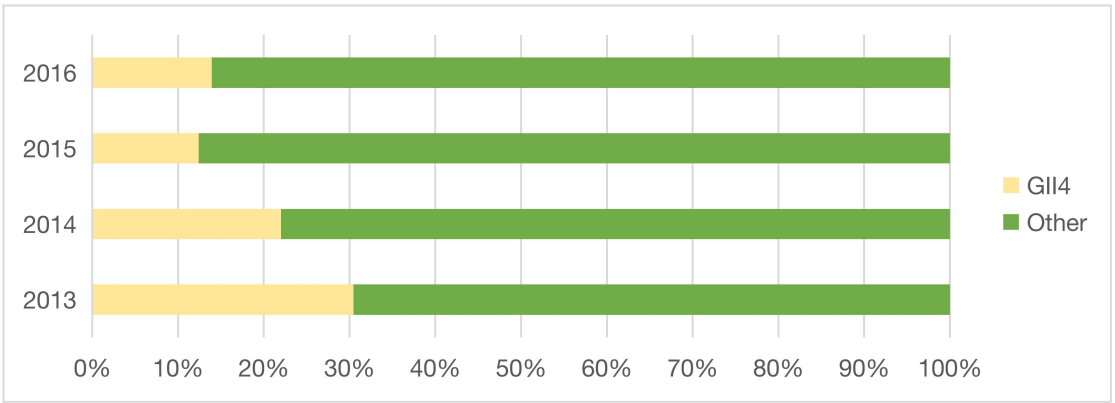

2016
2015
2014
2013
0%
10%
20%
30%
40%
50%
60%
70%
80%
90%
100%
GI1
GI2
GI3
GI4
GI5
GI6

(b)

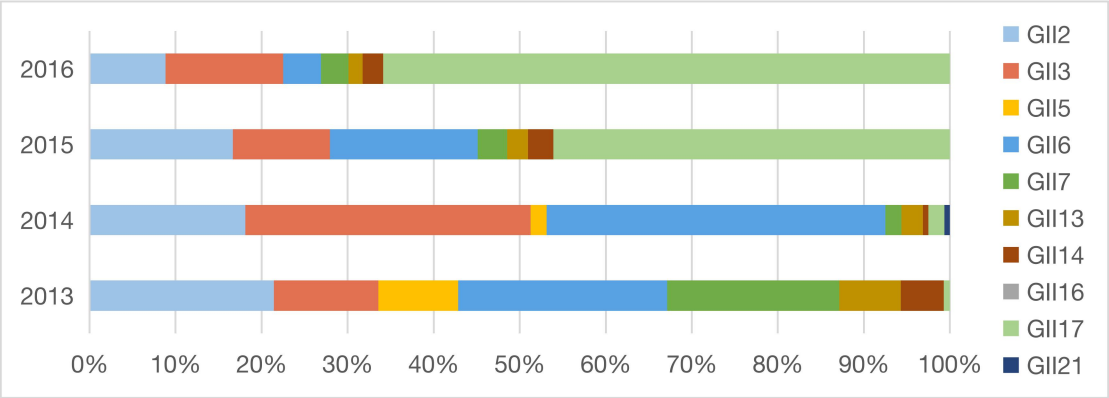

2016
2015
2014
2013
0%
10%
20%
30%
40%
50%
60%
70%
80%
90%
100%
GI1
GI2
GI3
GI4
GI5
GI6

(c)


2016
2015
2014
2013
0%
10%
20%
30%
40%
50%
60%
70%
80%
90%
100%
GII2
GII3
GII5
GII6
GII7
GII13
GII14
GII16
GII17
GII21


2016
2015
2014
2013
0%
10%
20%
30%
40%
50%
60%
70%
80%
90%
100%
GII4
Other

(d)



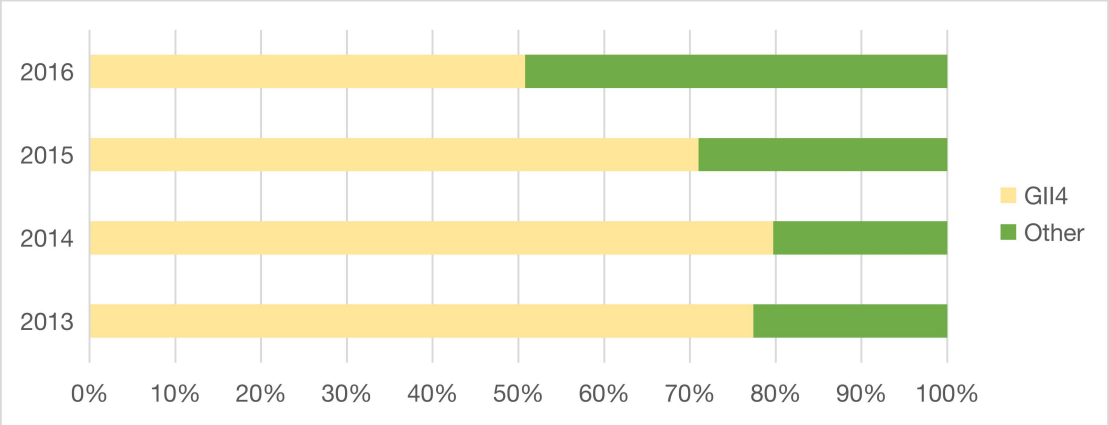


Figure S1 Distribution of NoV genotypes in wastewater(a)(c) and AGE cases[1](b)(d), 2013 ~ 2016. Comparing the temporal distribution of each genotype in sewage and in AGE cases, a similar trend can be observed for GII.

[1] Source: ORF2 sequences obtained for NoV outbreak investigations or the diagnosis of sporadic gastroenteritis cases from NoroNet (van Beek J et al. (2018): Molecular surveillance of norovirus, 2005–16: an epidemiological analysis of data collected from the NoroNet network. Lancet Infect Dis 18, 545-553 https://doi.org/10.1016/S1473-3099(18)30059-8)